20/1/26

Experiment -1

study of various Network commands used in Linux and Windows

1) Pathping google.com

Description: It traces the route to Google and analyzes each network hop for latency and packet loss to help diagnose network problems.

2) hostname - Displays the name of local computer on the network.
getmac - Displays the Mac address of the network adapters on the computer

3) tracert google.com.
Description: shows the path taken to reach google from your computer.

4. Route Print

Description: Displays the routing table of your computer.

5) ipconfig /flushdns

Description: clears the DNS cache stored on your computer.

nsloopup google.com

Description: finds the IP address of google.com using the DNS server.

6) ipconfig/release

Description : Release the current IP address assigned to your computer.

7) Ping www.google.com

Description : Checks if your computer can reach google and measures the response time.

ipconfig

Description: Displays the IP address and network configuration of your computer.

arp -a

Description displays the ARP table, showing the IP and MAC addresses

Experiment - 2

Learning and Assignment of IP Address manually to Computers

Aim:

To manually assign a static IP address to a computer and verify the network configuration.

Requirements (Inputs):

* Computer with windows OS
* Network connection (Ethernet/ Wifi)
* IP address: 192.168.100
* Subnet Mask : 255.255.255.0
* Default Gateway : 192.168.1.1
* DNS Server : 8.8.8.8

Procedure (Steps):

1) Open Control panel
2) Click Network and sharing Centre.
3) Select change Adapter settings
4) Right-click the active network adapter and choose properties
5) Select Internet protocol version 4 (TCP/IPV4) and click properties
6) choose Use the following IP address
7) Enter the following values.
   * IP address : 192.168.1.100
   * Subnet Mask : 255.255.255.0

* Default Gateway : 192.168.1.1

8) Under use the following DNS server address, enter:
   * Preferred DNS Server : 8.8.8.8
   * Alternate DNS Server : 8.8.4.4 (optional)
9) Click OK, then close
10) Open Command Prompt and type ipconfig to verify the assigned IP address.

Output

The computer is successfully configured with a static IP address, subnetmask, default gateway, and DNS server are displayed correctly, and the computer can communicate with other devices on the network.

Result :

The manual assignment of the IP address to the computer was completed successfully and the network settings were verified.

observation:

Hub

* Operates at the Physical layer
* Broadcasts data to all connected devices.

Switch

* Operates at the Data Link layer
* Uses MAC addresses to forward data.
* Sends data only to the destination device.
* Provides better speed and efficiency

To apology used in college

Star Topology

Advantages of star Topology

* Easy to install and manage.
* Easy Fault detection.

Comparison of Hub and Switch

Hub	Switch
* Physical Layer Device	* Data Link Layer Device
* More collission	* Less Collission
* No MAC address table.	* Maintains MAC address table

Experiment - 4

Experiment on CISCO Packet Tracer Analysis of Hub and Switch using Ciro Packet Tracer

Aim:

To analyze the behavior of Hub and switch by designing simple networks in Cisco Packet Tracer.

Procedure:

* Network using Hub
  1) Place one Hub and 4-6 PCs
  2) Connect all PCs to the Hub
  3) Assign IP addresses
  4) Send Ping Packets and observe communication.

* Network using switch
  1) Place One switch and 4-6 PCs
  2) Connect all PCs to the Switch
  3) Assign IP addresses
  4) Send Ping Packets and observe communication.

* Network Using Hub and switch
  1) Place one Hub, one switch and PCs
  2) Connect some PCs to the Hub and others to the switch.

* Place the connector in the Crimping tool and press firmly.
* Repeat the same process for the other end of the cable.

Output:

* Different types of network cables were identified and studied.
* The UTP cable was successfully ~~crip~~ crimped with RJ45 connectors
* The cable tester confirmed proper connectivity.

Result:

The different types of network cables were studied successfully. A UTP cable was crimped with RJ45 connectors using the T568B standard.

* Used in cable TV networks and older Ethernet networks.
* More resistant to interference than twisted pair cables.

3) Fiber Optic cable
* Transmitter data using light signals
* Offers very high speed and long-distance communication.
* Immune to electromagnetic interference
* Commonly used in internet backbones and high-speed networks.

Procedure:

* Study the different types of network cables : Twisted Pair, Coacial and fiber optic.
* Cut the UTP cable to the required length.
* Untwist the wire pairs and arrange them according to the T568B color code.

i) White/orange ii) Orange iii) White/Green
iv) Green v) White/Brown vi) Brown
vii) Blue.

* straighten and trim the wires evenly
* Insert the wires into the RJ45.

Experiment -3

Study of different types of Network Cables and crimping of cable with RJ45 connector

Aim:

To Study the different types of network cables and perform crimping of a UTP cable using a RJ45 connector.

Requirements (Input):

* UTP (cat5/cat6) cable
* RJ45 connectors.
* Crimping tool
* Wire stripper/cutter
* Cable tester (optional)

Types of Network cable

1) Twisted pair cable

* Most commonly used cable in LANs
* Contains pairs of copper wires twisted together to reduce interference.
* Types

UTP (Unshielded Twisted Pair)

STP (Shielded Twisted Pair)

2) Coaxial Cable

* Has a single copper conducter with insulation and shielding.

Result:

A network using Hub, switch and both devices together was successfully designed and simulated in Cisco Packet Tracer. The functioning differences between Hub and switch were observed, and the star Topology used in the college.

Output :

```
PS C:\Users\Admin\Desktop\ping> Python ftp_scanner.py

Enter FTP server address : ftp.gnu.org

Anonymous FTP SCANNER

Target : ftp.gnu.org

Port : 21

[1] connecting to FTP server...
[+] FTP server is reachable
[+] Server response: 220 GNU FTP server ready

[2] Testing anonymous login....
[+] Anonymous login successful
[!] The FTP server permits anonymous access

[3] checking accessible directory....
[+] Directory listing is available:
CRYPTO.README
find.txt.gz
gnu
non-gnu
old.gnu
pub
robots.txt
welcome.msg
```

Experiment - 6

Building an Anonymous FTP Scanner Using Python

Aim:

To develop a Python program that scans an FTP server and checks whether anonymous login is allowed.

Procedure:

1) Install Python and VS code
2) Create a Python program for FTP Scanning
3) Enter the FTP server address
4) Connect to the FTP Server
5) Check for anonymous login access
6) Display the result
7) Close the FTP connection.

Code:

```
from ftplib import FTP
from ftplib import all_errors
def anonymous_ftp_scan(host, port=21):
    print("=" * 50)
    print(f" Target : {host}")
    print("ANONYMOUS FTP SCANNER")
    print(f"Port : {port}")
    print()
ftp = FTP()
```

ing> py ping.py

: google.com

220.110] with 32

bytes=32 time=
13 ms
L=118
: bytes=32 time=3
ms
=118

32 time=3ms

L=118

es=32 time=3ms

=118

.220.110:

=4, Lost=0(0%
Loss),

in milli-seconds:

s ms,

=> Detect the operating system
=> Execute the appropriate ping command.
=> Display the ping result
=> check whether the server is reachable
=> Observe the output and verify Connectivity.

Code:

```
import subprocess
import platform
print ("=" *50)
print ("Customized Ping Utility")
print("=" * 50)
host = input ("Enter Website or IP address)
system = platform.system ()
if system == "Windows"
    command = ["ping", "-n", "4", host]
else:
    command = ["ping", "-c", "4", host]

try:
    result = subprocess.run(command, capture-
                output = True, text = True)
    print ("\nPing Results\n")
    print(result.stdout)
    if result.returncode == 0:
        print("Server status: Reachable")
    else
        print("Server status: Unreachable")
except Exception as e:
        print("Error:", e)
```

Result: Thus, a customized Ping command was successfully developed and executed.

Output:

PS c:\Users\Adwin\Desktop\ping> py ping.py

customized ping Utility

Enter website or IP address : google.com

Ping results

Pinging google.com [142.251.220.110] with 32 bytes of data

Reply from 142.251.220.110: bytes=32 time=13 ms TTL=118

Reply from 142.251.220.110: bytes=32 time=3ms TTL=118

Reply from 142.251.220.110: bytes=32 time=3ms TTL=118

Reply from 142.251.220.110: bytes=32 time=3ms TTL=118

Ping statistics for 142.251.220.110:

Packets: Sent=4, Received=4, Lost=0(0% Loss),

Approximate round trip times in milli-seconds:

Minimum=3ms, Maximum=13 ms,

Average=5 ms

Server status: Reachable

Experiment - 5

Aim:

To develop a customized ping command in Python that checks the connectivity between a client and a server by sending ICMP Echo Request packets and displaying the response time and status.

Requirement:

Hardware
=> Computer / Laptop
=> Internet Connection

Software
=> Windows 10/11 or Ubuntu Linux
=> Python 3.10 or above
=> Visual Studio Code or pycharm
=> Command prompt / Terminal

Theory:

A Ping command is a network diagnostic tool used to determine whether. It works using the ICMP
=> Enter any hostname or IP address
=> Send multiple ping requests
=> Display response time
=> Accept a hardware or IP address.

using

on a network
scans ARP

, UDP ping
hich protocol.

urce network
ndem Lyon

open the
o om

and answer

and start.
bnet.

4 record

TMP

6) Run TCN SYN (PS) TCP ACK, UDP scan.

7) Complete the knowledge check and proceed to Nmap Base Port Scan room.

Result:

Thus nmap host discovery scan were performed successfully using TCP, SYN, FCP, UDP ping technique the scan result network traffic were observed.

Experiment -8

NMAP to Discover Live Hosts using NMAP scans

Aims :

To discover live ~~lost~~ host on a network using Nmap host discovery scans ARP scans. ICMP Scam and TCP, UDP ping scan - and to understand which protocol

Nmap :

Nmap is a free open-source network scanning tool created by gordem Lyon and first released.

Procedure :

1) Log into Tryttack and open the Nmap live Hosts Discovery room
2) Work through tasks 1-3 and answer the simulator based question.
3) Deploy lab network Task 4 and start attackbox note target Subnet.
4) Run ARP scan. PR <subnt >/24 record many host responed.
5) Run ICMP timestamp and IMP address.

sender buffer

Frame No: 0 Data: M

Frame No: 1 Data: E

Frame No: 2 Data: L

Receives response: Act 3

all frames transmitted successfully!

original message: Hello

Receiver code:

```
import os
SENDER-FILE = "Sender_buffer.text"
RECEIVER-FILE= 'reciever_buffer.txt"
expected frame = 0
while true:
    if not os.path.exist(sendor-file);
    print( In framer received )
     error =false
     error_frame = none
```

Result:

Thus, the sliding window protocol was implement successfully using python and flow of frames Sender and receiver was simulated ACK/NACK.

Experiment 7

Sliding window protocol

Aim:

To write a program to implement flow control at data limit layer using sliding window protocol stimulate the flow of frames from one node to another.

Sender code:

```
import time
print("="*55)
Print("Sliding window protocol stimulate sender")
window-sizel =int(input("Enter window size"))
if(window_size<=0):
    Print("Invalid window size!")
    exit()
message = input("Enter text message:")
if not message:
    Print("message cannot be empty!")
    exit()
frames=[]
for i, character in enumerate(message):
    framed_append({"frames_no"}
```

```
try:
    print("[1] Connecting to FTP server...")
    ftp.connect(host, port, timeout=10)
print ["[+] FTP server is reachable")
print ["[+] Server response:", ftp.get
                                welcome())
print ("\n[2] Testing anonymous login...")
ftp.login(user="anonymous", password=
                        "anonymous@example.com")
print ("[*[+] Anonymous login SUCCESSFULL")
print("[!] The FTP server permits anonymous
                                access.")
print ("\n[3] checking accessible
                                directory...")
files = ftp.nlst()
if files:
    print("[+] Directory listening is
                                available.")
    for item in files:
        print("  ", item)
else:
    print("[-] Directory is empty or
                                listing is unavailable.")
```

Result:

Thus, an Anonymous FTP ~~server~~ scanner was successfully developed using Python and tested for anonymous FTP access.

Experiment - 11

Routing at Network Layer

Aim:

To stimulate static routing and RIP using cisco Packet tracer.

A) Static Routing:

Static routing is method where routes are manually added to the router routing table. It is useful for controlling path taken by packet biff network.

Procedure:

1) open cisco packet tracer.
2) Place required routers, switch and PC in workspace.
3) Connect the device using suitable cables.
4) Assign IP address to all PCs and router interfaces.
5) Open the router CLI
6) Enter the full command to configure static route.

Procedure:

1) Open cisco packet tracker
2) place one router and two PCs in workspace.
3) Connect PCs to reader 1 FastEthernet using copper straight through cable.
4) Connect PC, to router FastEthernet 011 using copper Straight through cable.
5) Open route CLI and enter.
6) Configure PC0:

IP: 192.168.10.2
Subnet mask: 255.255.2550
Default Gateway: 192.168.10.1

7) observe the reply packet.
8) Send a simple PD from PC0 to PC1 and observe the packet transmiss oin.

Result: Thus internork was successfully designed and configured using a router in cisco packek tracer.

Experiment -10

Internetworking with routers in cisco packet tracker simulator.

Aim:

To design and configure a simple internetwork using router and establish communication.

Requirements:

* cisco packet Tracker
* Router
* PCs
* copper straight through cables.

Theory:

A router is a networking device used to connect different network and forward Packet between them. In two PCs connected to two different interface of routes static IP address assigned in PCs and router inferface.

Example commands:

Enable

Configure terminal

IP address 192.168.1.1.255.255.224

no shutdown

exit

5) Check the connection between PCs using Ping command.

6) Send a simple PDU between PCs in different subnet.

Result:

Thus, subnetting was successfully implemented in cisco packet tracer.

Experiment - 9

Implementation of subnetting in cisco packet Tracker simulator.

Aim:

To implement subnetting in cisco packet tracker by creating multiple subnets, assigning static IP address connecting the subnets.

Requirments:

* cisco packet tracer
* Router
* Switches
* PCs
* Copper Straight-through cables.

Procedure:

1) Open ~~source~~ cisco packet tracer and create a new network topology.
2) Add the required router, switches and PCs to workspace
3) Connect PCs to switch to router using suitable cables.
4) Divide the network 192.168.1.0/24 smaller subnet using 127 subnet mask.

Experiment 13

Implement your own ping program

Aim:

To implement a simply ping Program to check connectivity two network.

Theory:

Ping is a network utility used to check whether a destination device is resource network. It sends a request to destination and waits for reply.

Requirments:

* Compiler
* Python/suitable programming.
* Network connection.

Procedure:

1) Create a new program for implementation ping operation.
2) Import required socket/network library.
3) Get IP address or hostname of the destination.

b) Client servers using socket progra

1) Create server and client program using socket programming.

2) start servers and bind it Ip address and port.

3) Make sowers wait for client.

4) The server sends reply back to client

5) Continue sending and receiving msg to perform a chat.

Result:

The and to end communication was succesfully implemented using TCP/UDP sockets.

Experiment - 12

End-to-End communication at transport layer.

Aim:

To implement an echo client server using TCP / UDP socket programming.

a) Echo - client server using TCP / UDP a socket is an endpoint used for communication between two devices. In echo program, the client sends a message.

Procedure:

1) Create server program, client program
2) Import the required socket library
3) In server, create a socket and bind it and IP address.
4) Make the server listen to client request.
5) start client and connect it sever the client receives and display.

Observation:

In static routing, router are manually configured by administor

In RIP, router automatically exchange routing information. RIP selects a route based on lowest hop count.

Result:

Thus, static routing and RIP were successfully simulated using Cisco packet Tracer.

7) Usefull command to check routing
show IP route.

B) RIP routing :

RIP is a dynamic routing protocol.
It allows routers to automatically.
Exchange information abt the network
connected to them.

Procedure :

1) Create network with three
routers interfaces.

2) Assign IP address to pcs and
router interfaces.

3) Configure the route interface usin
-g.

4) Repeat RIP configuration for other
routers with their directly connected

5) Repeat RIP configured for other
router with their directly connected
network.

6) Allow some time for routers to
exchange.

7) send simple PDV from source.
PC to destination Pc.

Input:
A web sever log file

Output:
Web log analysis report generated using webalizer tool.

Result:
Thus, the web log file was successfully analyzed using webalizer tool.

Experiment -15

Analysis of web logs using Webalizer tool

Aim:

To analyse the diff types of web logs using webalizer tool.

Theory:

Webalizer is a web log analysis tool used to process web server.

Procedure:

1) Run the webalizer window
2) Select or provide the required web log.
3) Download web log file from web if required
4) Load the web log file into
5) Study the diff information available.

Procedure:

* Open Python IDE or Terminal
* Create a python program for Packet sniffing.
* Import socket module
* Use recvfrom() to receive packet.
* Generate some network traffic and observe the captured packet.

Result:

Thus, a RAW socket based Packet filling program was implemented.

Experiment - 14

Raw sockets - Packet sniffing

Aim:

To write a program using RAW sockets to implement packet sniffing.

Theory:

A packet sniffer is a program used to capture and examine network packets.

Requirement:

Computer / laptop

Python

RAW socket support

Network interface.

Algorithm:

~~Computer / laptop~~

1) Start the program

2) Create RAW socket

3) Find the socket required network interface.

4) Wait for incoming network socket.

5) Capture the packet using RAW

4) Send packet to destination
5) If response is received, display destination.
6) Display the result and close the connection.

Algorithm:

1) start
2) Entire destination IP address
3) Create a socket
4) Send ping request
5) wait for response.
6) otherwise display host is unreachable.

Result:

Thus, the ping program was successfully implemented.